

# **Application of Information & Communication Technologies**

## **Lecture-28**

# Overview of Lecture 28

- ◆ Revise for Final

# Databases

- A collection of related data organized for quick access and retrieval.
- Stores data in tables (rows and columns).
- Example:
  - Student records, bank transactions, library catalog.

# Data, Information, and Tables

- Data: Raw facts (e.g., “Ali”, “21”, “CS”).
- Information: Processed data with meaning (e.g., “Ali is 21 years old”).
- Table: A structured collection of data (rows = records, columns = fields).
  - sample student table

| StudentID | Name  | Age | Department | GPA |
|-----------|-------|-----|------------|-----|
| 101       | Ali   | 21  | CS         | 3.6 |
| 102       | Sara  | 22  | AI         | 3.8 |
| 103       | Bilal | 20  | SE         | 3.4 |
| 104       | Hina  | 23  | IT         | 3.9 |

# Database vs File System

## File System

Stores files independently

Manual data linking

No query facility

Difficult to maintain

## Database

Stores related data together

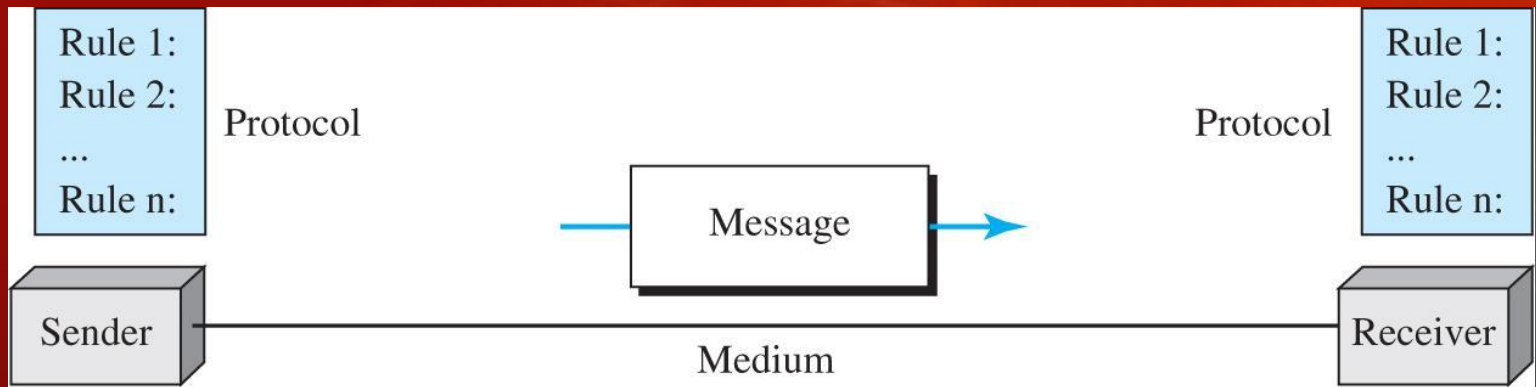
Relationships managed automatically

SQL-based queries

Easier to manage and secure

# Components of a Communication System

- ◆ A data communications system has five components:



# Components of a Communication System

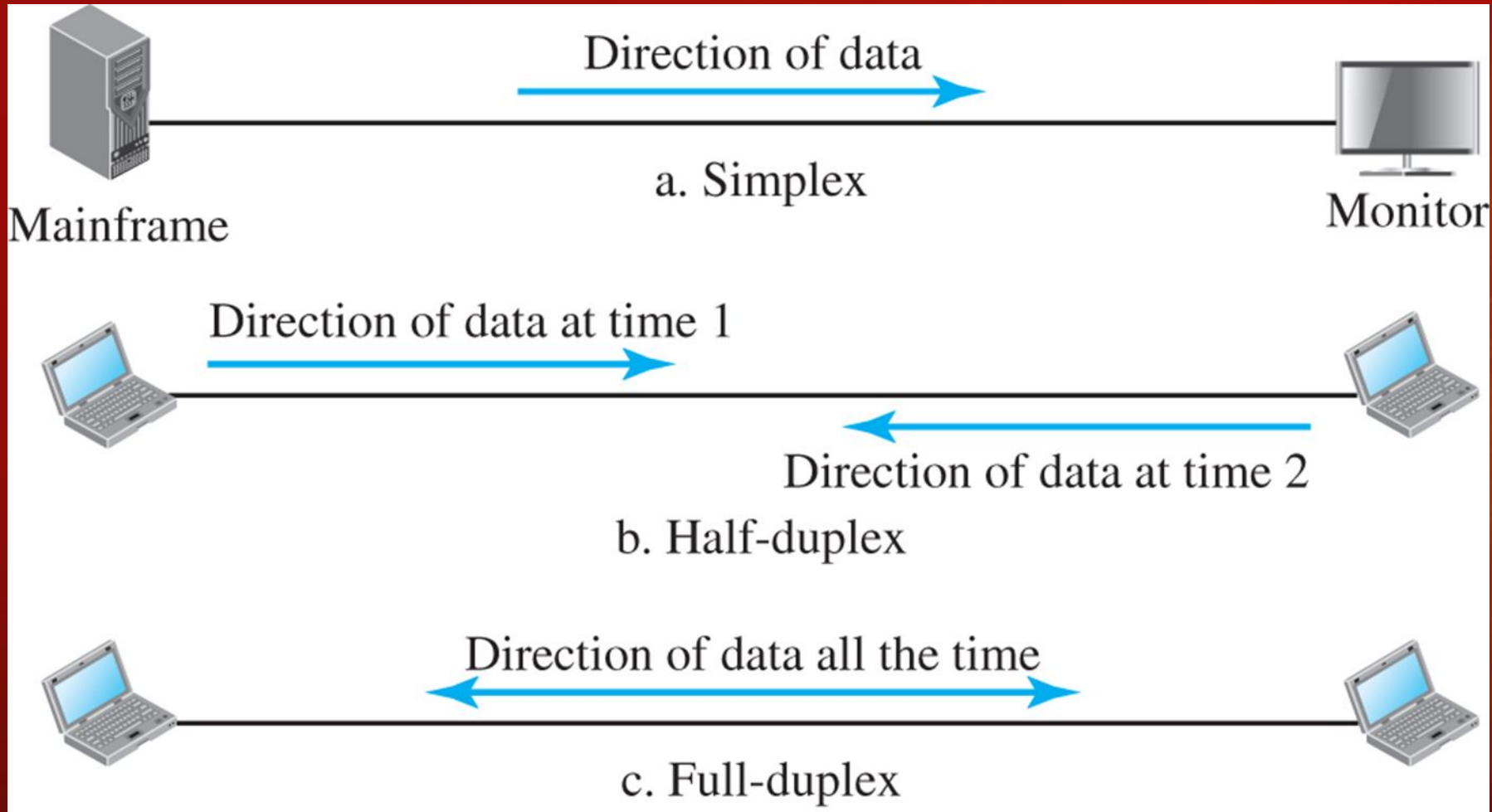
- ◆ A data communications system has five components:
  - **Message:**
    - The actual information or data to be transmitted.
    - Example: text, audio, video, or file.
  - **Sender**
    - The device that creates and sends the message.
    - Example: computer, mobile phone, or sensor.
  - **Receiver**
    - The device that receives and interprets the message.
    - Example: server, printer, or another computer.
  - **Transmission Medium**
    - The path through which data travels.
    - Example: wired (fiber optic, cable) or wireless (radio, Wi-Fi).
  - **Protocol**
    - A set of rules that govern data communication.
    - Defines how data is formatted, transmitted, and received.
    - Example: TCP/IP, HTTP, FTP.

# Communication Modes

- ◆ Also known as Data Flow
- ◆ Communication between two devices can be simplex, half-duplex, and duplex



# Communication Modes

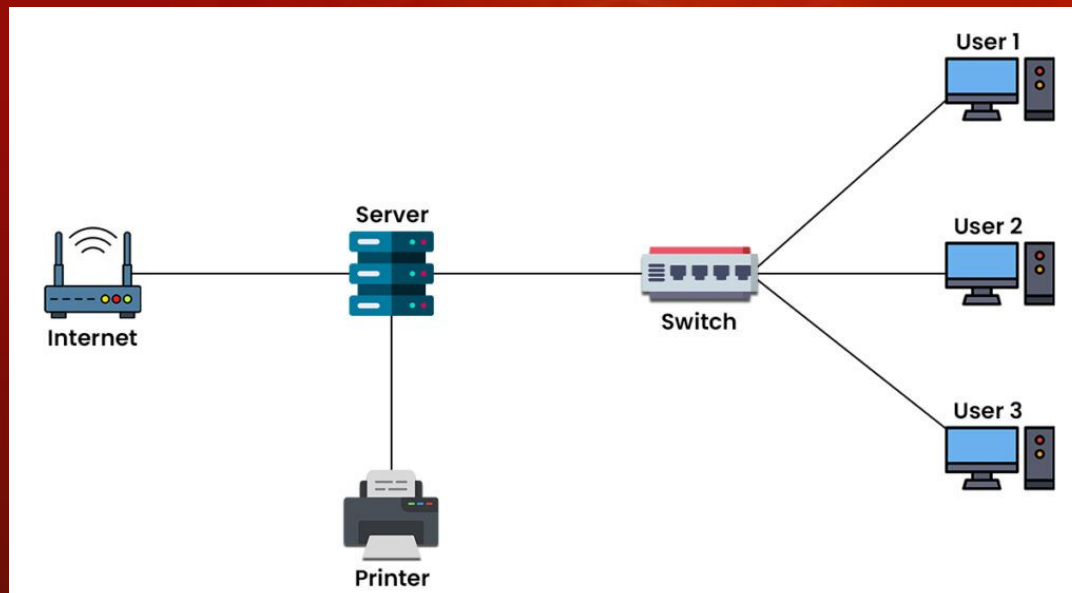


# Comparison of Data Transmission Modes

| Feature                | Simplex                                       | Half-Duplex                                                     | Full-Duplex                                    |
|------------------------|-----------------------------------------------|-----------------------------------------------------------------|------------------------------------------------|
| Direction of Data Flow | One-way only                                  | Two-way, but one direction at a time                            | Two-way, simultaneously                        |
| Communication Example  | Keyboard → Monitor                            | Walkie-Talkie                                                   | Telephone Call                                 |
| Transmission Control   | Only sender transmits; receiver only receives | Each device can both send and receive, but not at the same time | Both devices send and receive at the same time |
| Speed / Efficiency     | Slow (limited)                                | Moderate                                                        | Fast and efficient                             |
| Cost / Complexity      | Simple and inexpensive                        | Moderate                                                        | More complex and expensive                     |
| Use Case Examples      | Display units, sensors                        | CB radio, data link between two stations                        | Video conferencing, modern LANs                |

# Computer Network

- A Network is simply two or more computers that are linked together.
- A network is a group of connected computers and devices that share data and resources.
- Purpose: Share files, internet, printers, and applications.

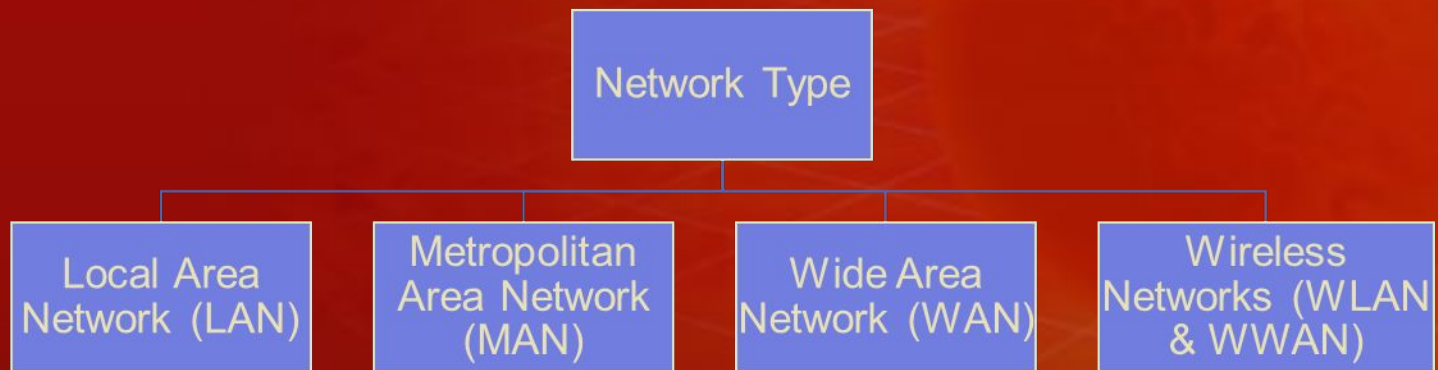


# Why Networking?

- ◆ Enables **communication** between users.
- ◆ Facilitates **data sharing** and resource utilization.
- ◆ Reduces **cost** by sharing devices.
- ◆ Enhances **collaboration** and remote work.

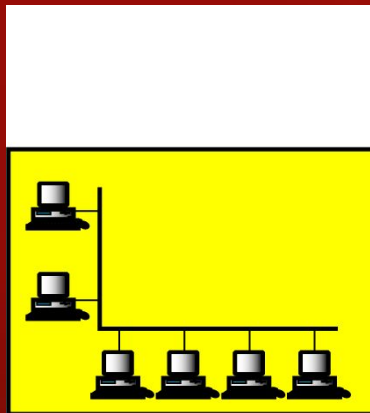
# Classification of Networks

- ◆ Networks can be categorized by size and coverage area:

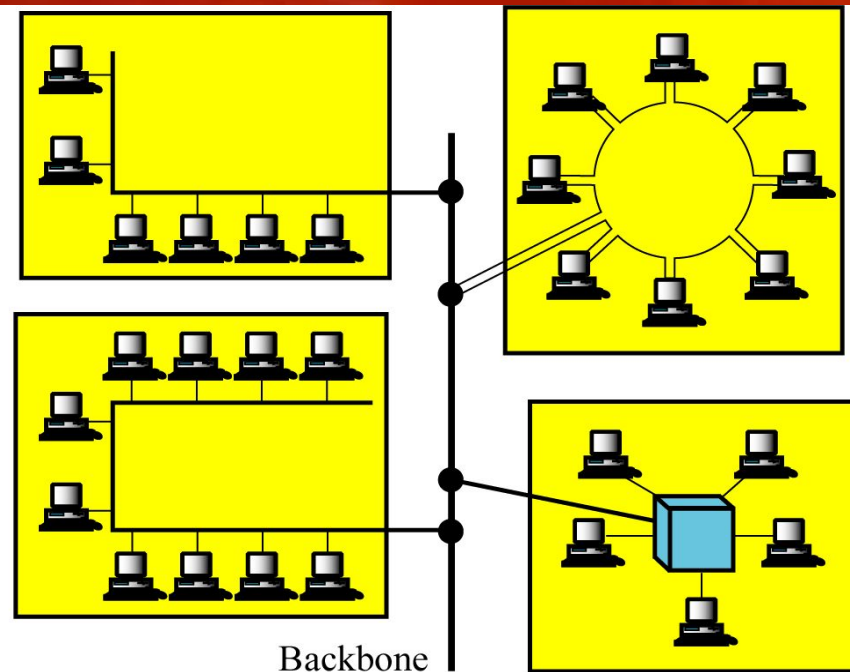


# Local Area Networks

- ◆ Covers a **small area** such as a room, building, or campus.
- ◆ **High speed** (up to 1 Gbps or more).
- ◆ **Privately owned** and maintained.
- ◆ Common in **schools, offices, and labs.**



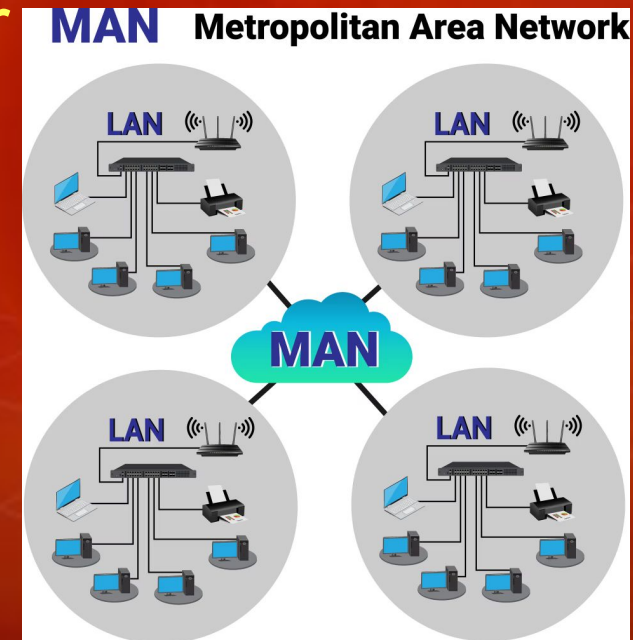
a. Single building LAN



b. Multiple building LAN

# Metropolitan Area Network (MAN)

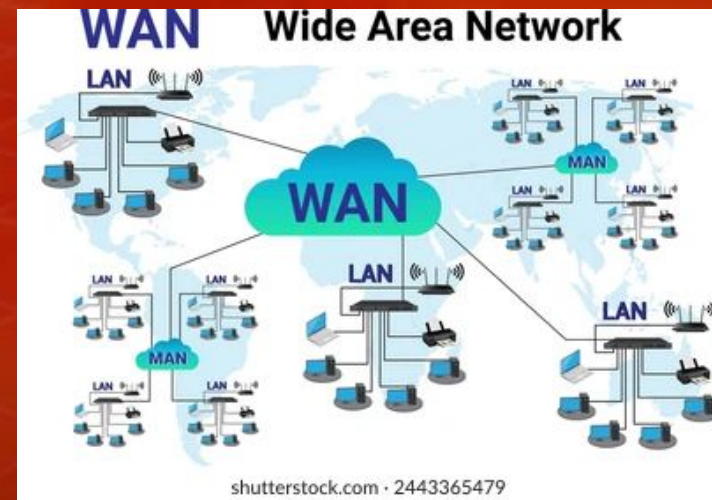
- ◆ Covers a city or large campus.
- ◆ Connects multiple LANs using fiber optics or wireless.
- ◆ Managed by ISPs or city authorities.
- ◆ Example: city-wide Wi-Fi or university campus network.





# Wide Area Network (WAN)

- ◆ Covers large geographical areas (countries or continents).
- ◆ Connects multiple MANs or LANs.
- ◆ Uses telecommunication lines, satellites, or undersea cables.
- ◆ Example: the Internet is the largest WAN.





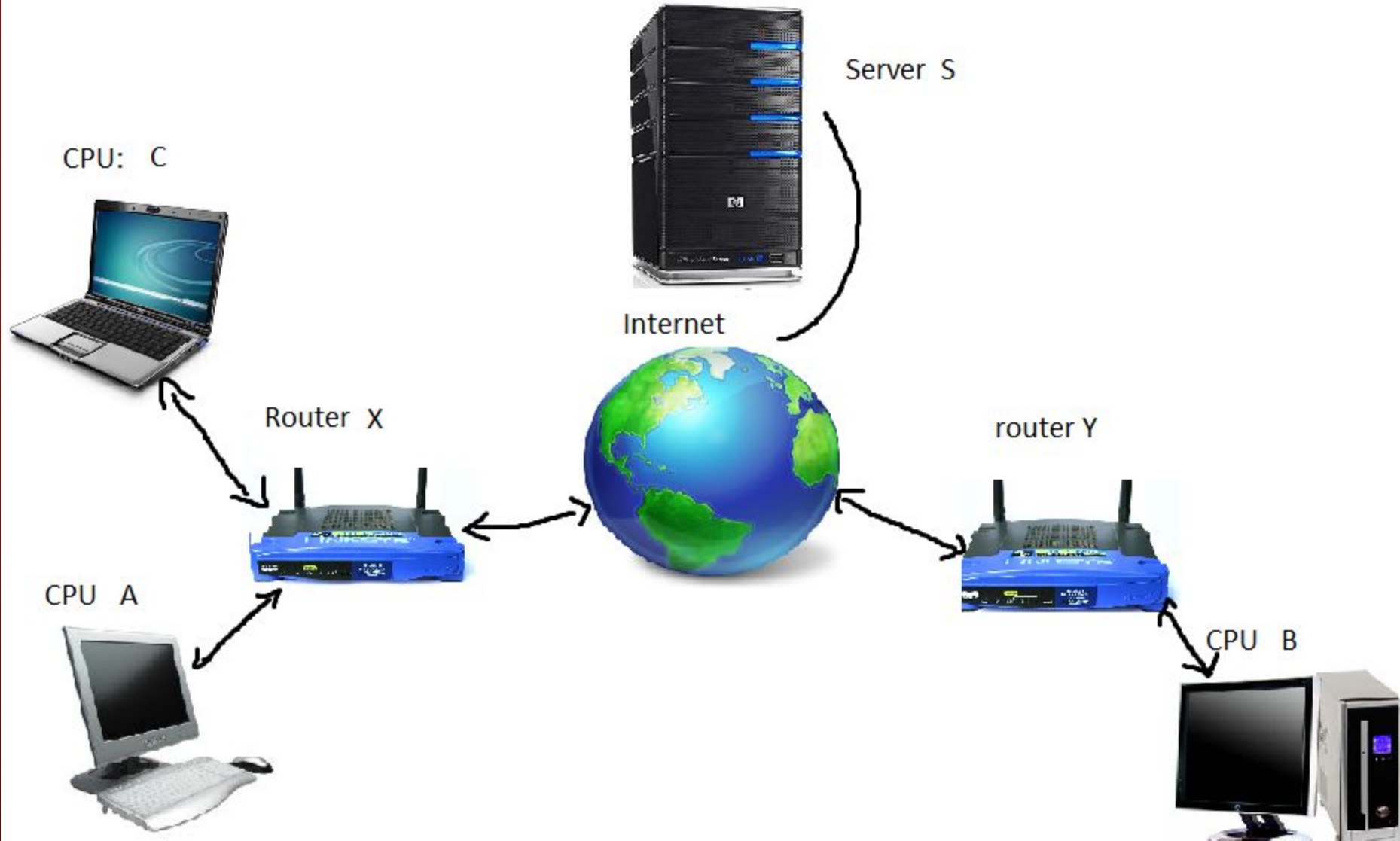
# Comparison Summary Table

| Network Type | Full Form                   | Coverage Area     | Speed    | Example            |
|--------------|-----------------------------|-------------------|----------|--------------------|
| <b>LAN</b>   | Local Area Network          | Building / Campus | High     | Office, School     |
| <b>MAN</b>   | Metropolitan Area Network   | City / Campus     | Medium   | University Network |
| <b>WAN</b>   | Wide Area Network           | Country / Globe   | Variable | Internet           |
| <b>WLAN</b>  | Wireless Local Area Network | Room / Office     | High     | Wi-Fi Network      |
| <b>WWAN</b>  | Wireless Wide Area Network  | Large Region      | Medium   | 4G/5G Mobile Data  |

# Client–Server Model

- ◆ A central server provides services or data to client computers.
- ◆ Clients send requests, and the server responds.
- ◆ Used in offices, banks, and web systems.
- ◆ Examples:
  - Web servers, email servers, database servers

# Client-Server Model



# Advantages & Disadvantages (Client–Server)

## Advantages

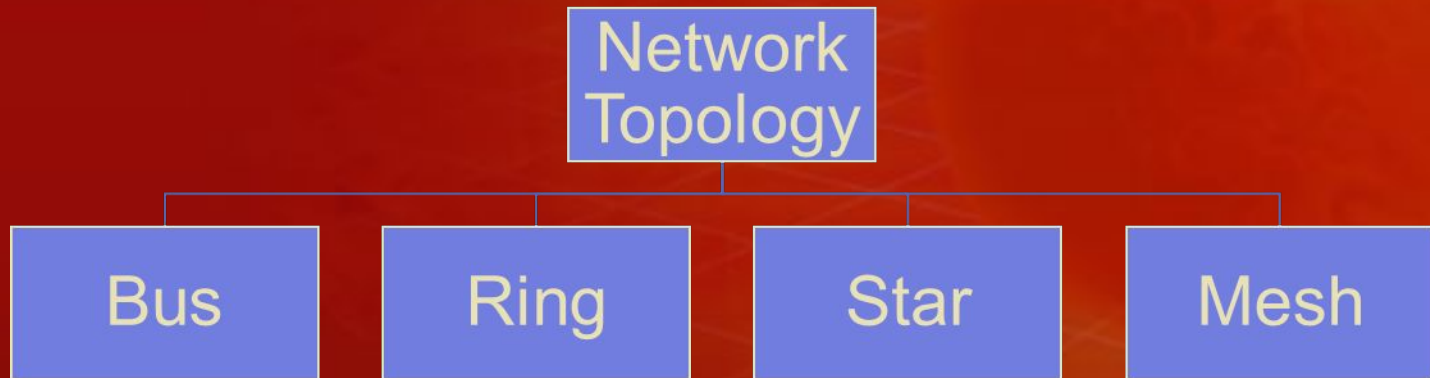
- Centralized control
- Easier data backup
- High security

## Disadvantages

- Server failure affects all clients
- Expensive to set up
- Requires skilled admin

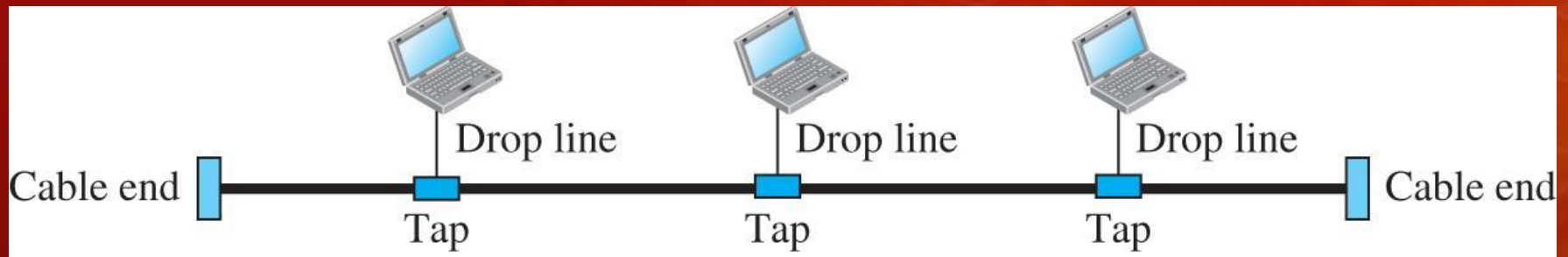
# Network Topologies Overview

- ◆ Network topology is the arrangement or layout of connected devices.



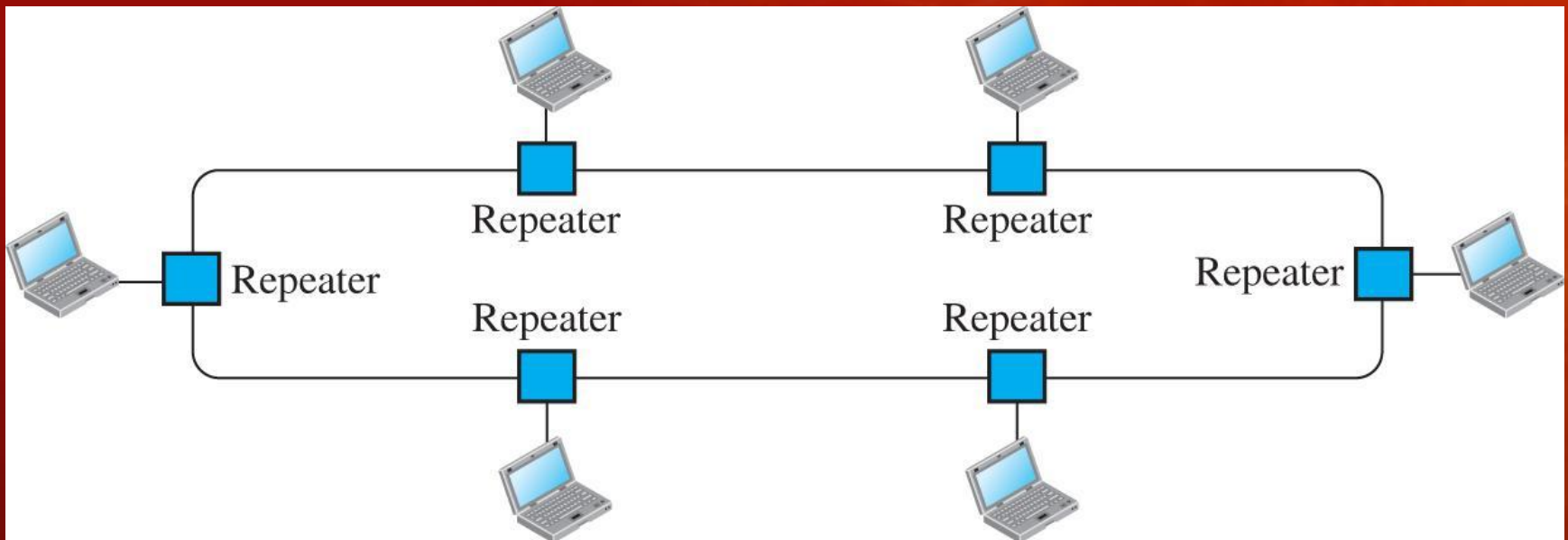
# Bus Topology

- ◆ All devices connected to a single backbone cable.
- ◆ Cost-effective
- ◆ Data collision possible, limited length



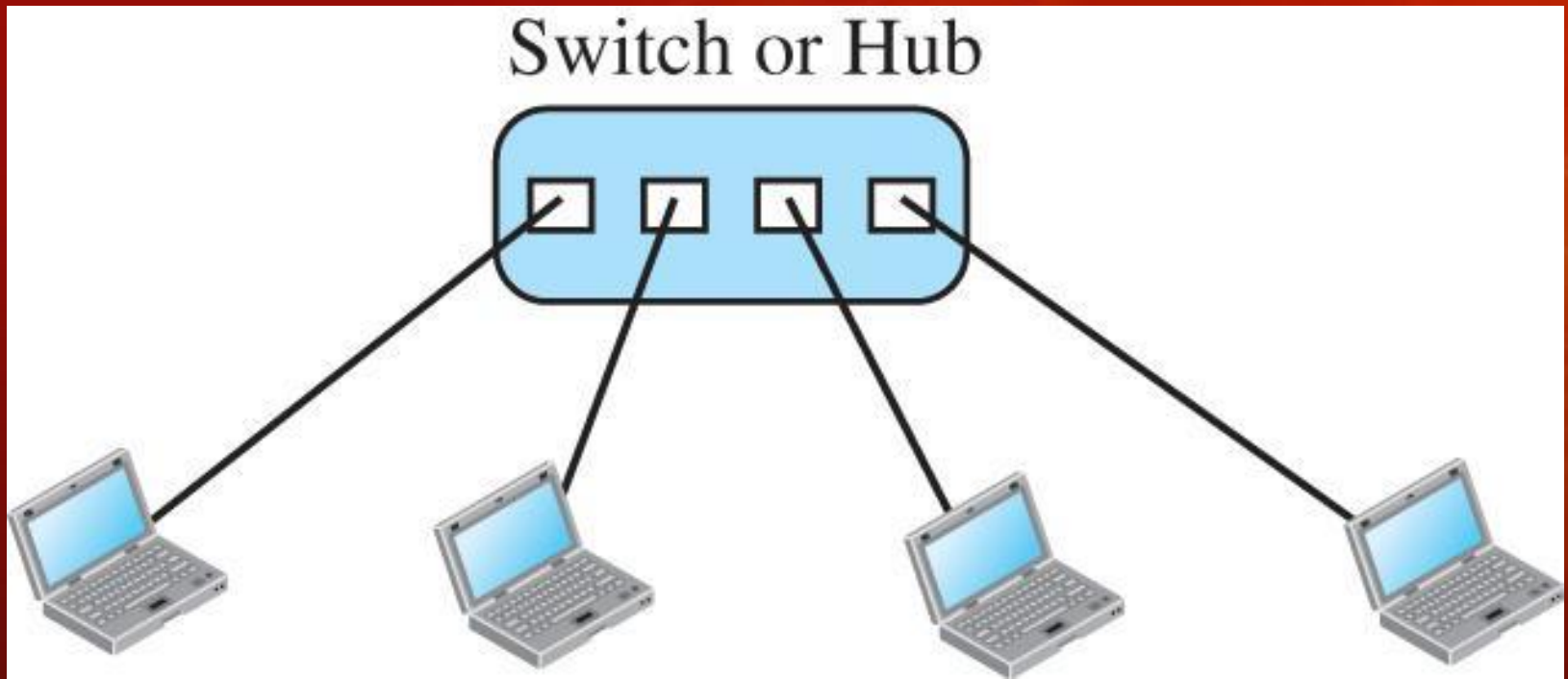
# Ring Topology

- ◆ Devices connected in a closed loop.
- ◆ Data travels in one direction.
- ◆ Easy to install
- ◆ One failure can affect the whole network



# Star Topology

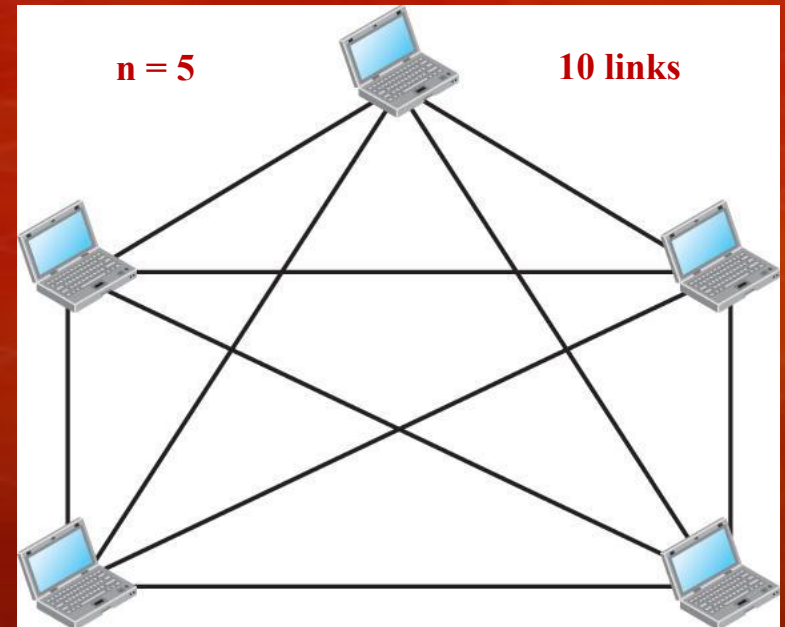
- ◆ All devices connect to a central hub or switch.
- ◆ If one cable fails, others continue working.
- ◆ Reliable, easy to manage
- ◆ If hub fails, network fails





# Mesh Topology

- ◆ Every device has a **dedicated** point-to-point link to every other device.
- ◆ Highly reliable and fault-tolerant
- ◆ Expensive and complex
- ◆  $n(n-1)/2$  in duplex-mode links.
  - $n=5$ , link=10

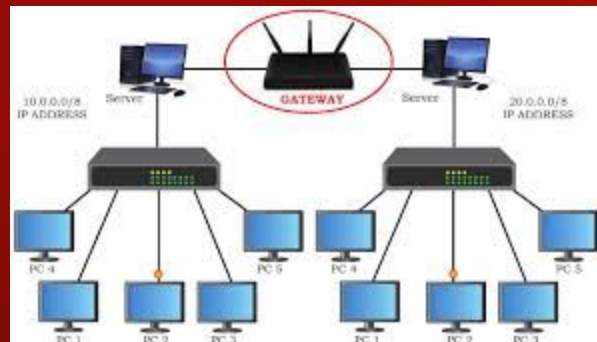


# Comparison Table of Topologies

| Topology | Main Feature    | Advantage         | Disadvantage            |
|----------|-----------------|-------------------|-------------------------|
| Star     | Central hub     | Reliable          | Hub failure             |
| Ring     | Circular path   | Orderly data flow | One failure affects all |
| Bus      | Common cable    | Cheap             | Collisions              |
| Mesh     | Fully connected | Fault-tolerant    | Costly                  |

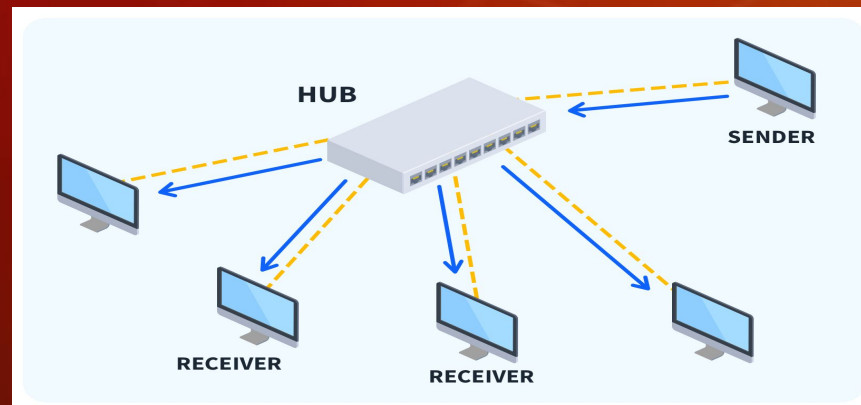
# Networking Devices

- Tools that connect, control, and manage communication between computers in a network.
- They enable data transfer, internet access, and network management.



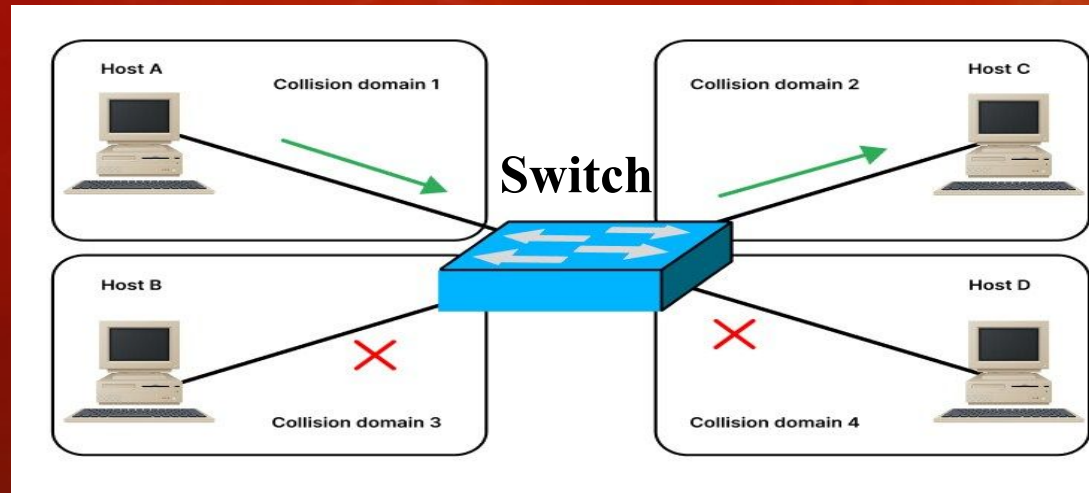
# Hub

- ◆ A Hub is a simple network device that connects multiple computers in a LAN (Local Area Network).
- ◆ It sends data received from one device to all other connected devices, regardless of the destination.
  - Broadcasts
- ◆ **Function:**
  - Works at the Physical Layer (Layer 1) of the OSI model
  - Does not filter or manage traffic
  - Shares bandwidth among all connected devices



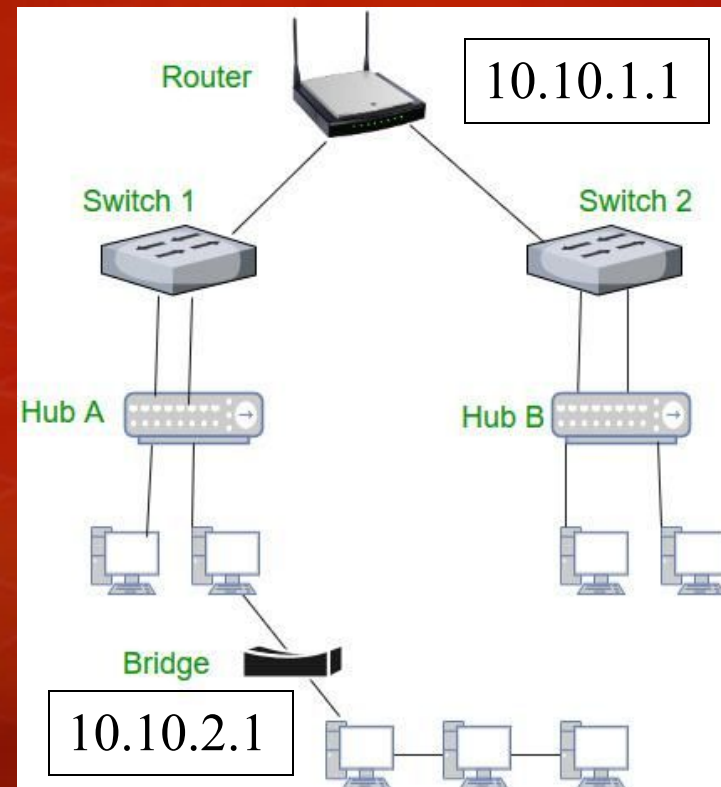
# Switch

- ◆ A device that connects multiple computers within a LAN and uses MAC addresses to send data only to the correct device.
- ◆ Function:
  - Improves network efficiency
  - Reduces data collisions



# Router

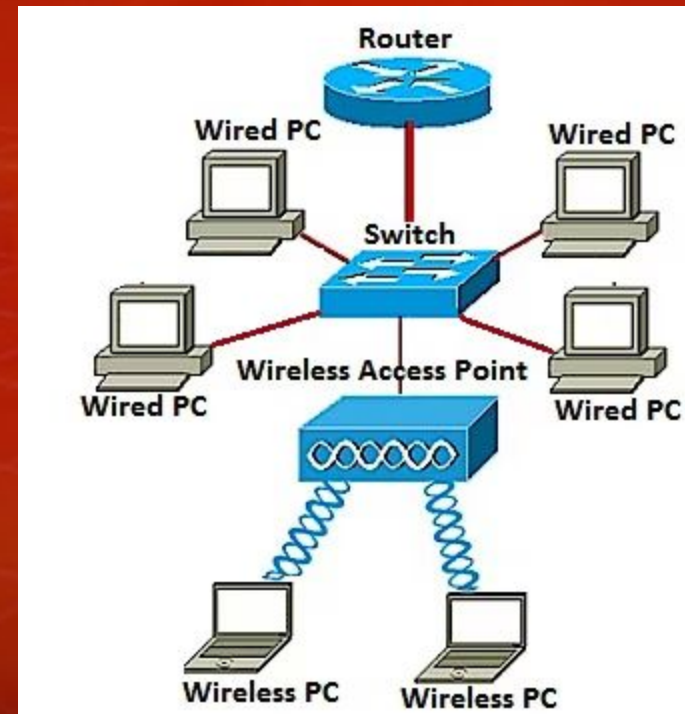
- ◆ Forwards data packets between networks based on IP addresses.
  - IPv4 is four byte address
  - 10.10.1.1 or 10.10.2.1
- ◆ **Function:**
  - Connects multiple networks (e.g., LAN to Internet)
- ◆ Manages traffic using routing tables
- ◆ Often includes a built-in firewall and Wi-Fi feature



# Access Point (AP)



- ◆ Extends the range of a wired network by allowing wireless devices to connect.
- ◆ Function:
  - Acts as a bridge between wired and wireless networks
  - Common in schools, offices, and large buildings





# Summary Table

| Device                       | Purpose                                         | Connection Type  | Main Function                                                         |
|------------------------------|-------------------------------------------------|------------------|-----------------------------------------------------------------------|
| Hub                          | Connects multiple computers in a LAN            | Wired            | Broadcasts data to all connected devices without filtering            |
| Switch                       | Connects devices within a LAN                   | Wired            | Sends data only to the intended device using MAC addresses            |
| Router                       | Connects different networks together            | Wired / Wireless | Directs (routes) data packets using IP addresses                      |
| Modem                        | Connects LAN to Internet Service Provider (ISP) | Wired            | Converts digital data ↔ analog signals                                |
| Access Point (AP)            | Provides wireless connectivity                  | Wireless         | Extends wired network for Wi-Fi-enabled devices                       |
| Network Interface Card (NIC) | Connects a computer to a network                | Wired / Wireless | Converts data into signals for sending and receiving over the network |

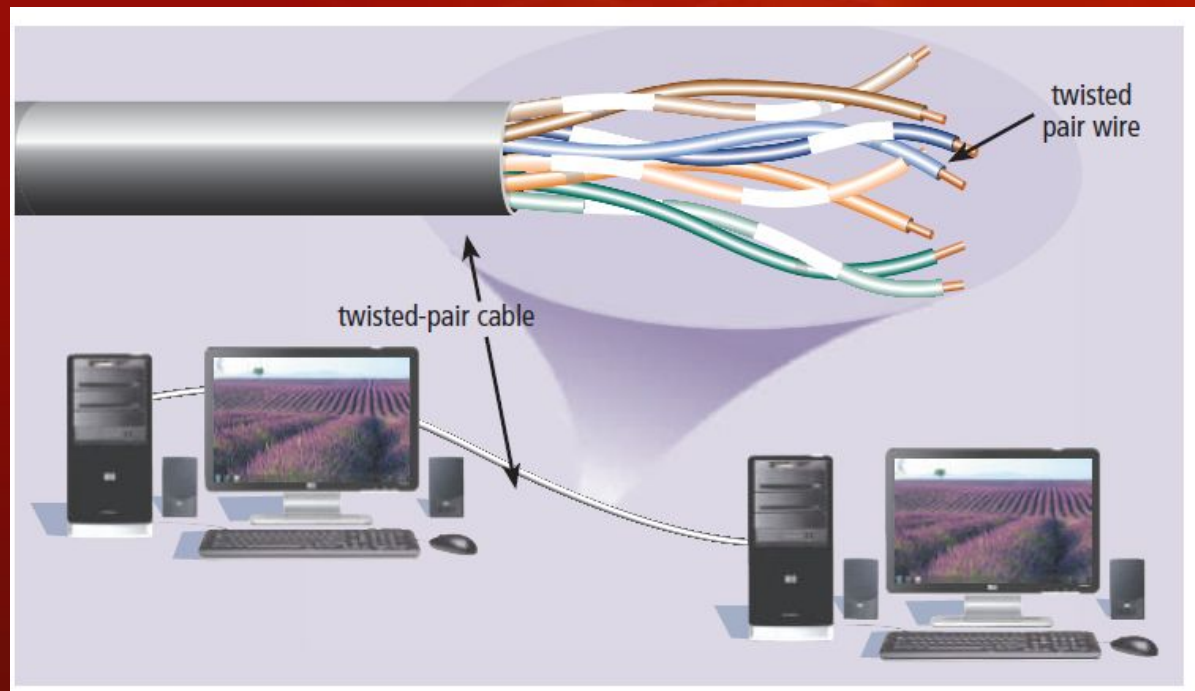


# Wired Media Overview

- ◆ Uses physical cables for data transmission.
- ◆ High security and reliability.
- ◆ Limited by distance and installation cost.

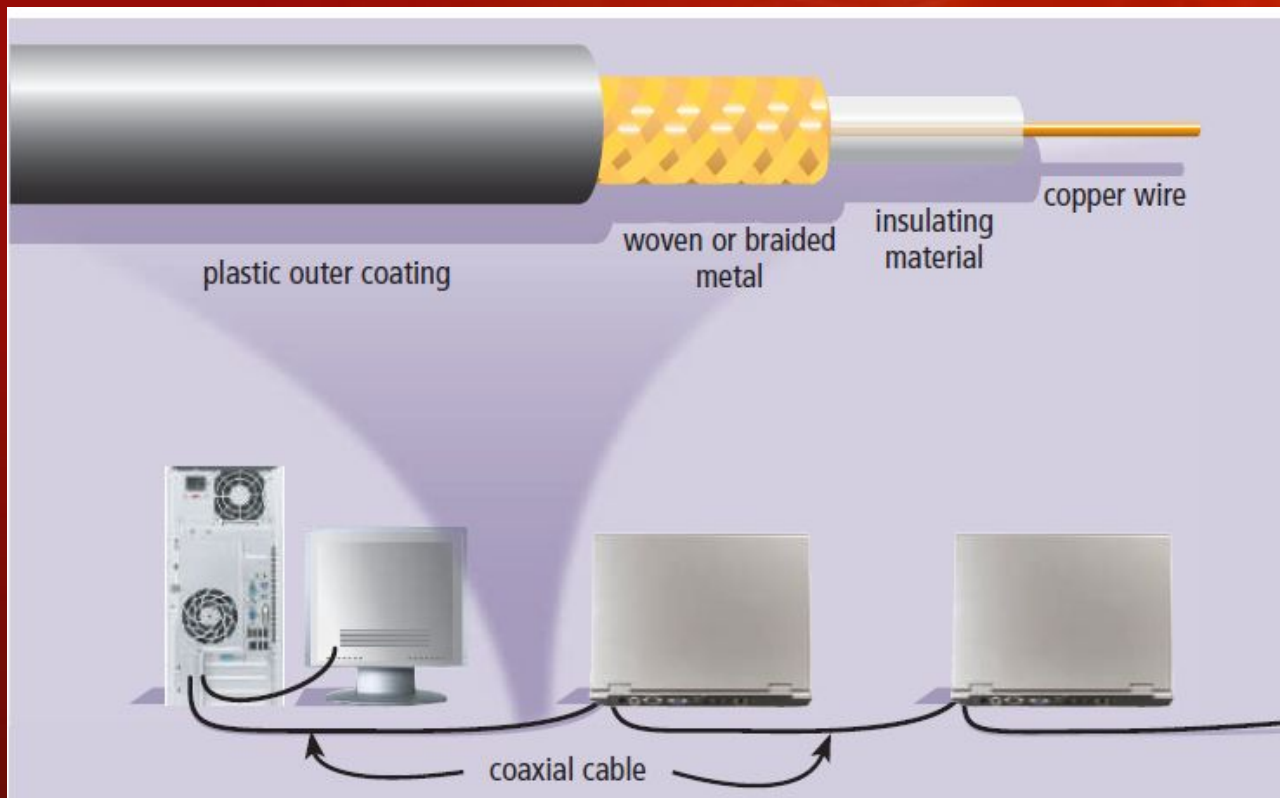
# Twisted Pair Cable

- ◆ Two copper wires twisted together to reduce interference.
- ◆ Common in telephone and LAN networks.
- ◆ Types:
  - Unshielded Twisted Pair (UTP) – cheaper, used in LAN.
  - Shielded Twisted Pair (STP) – less noise, used in industrial settings.



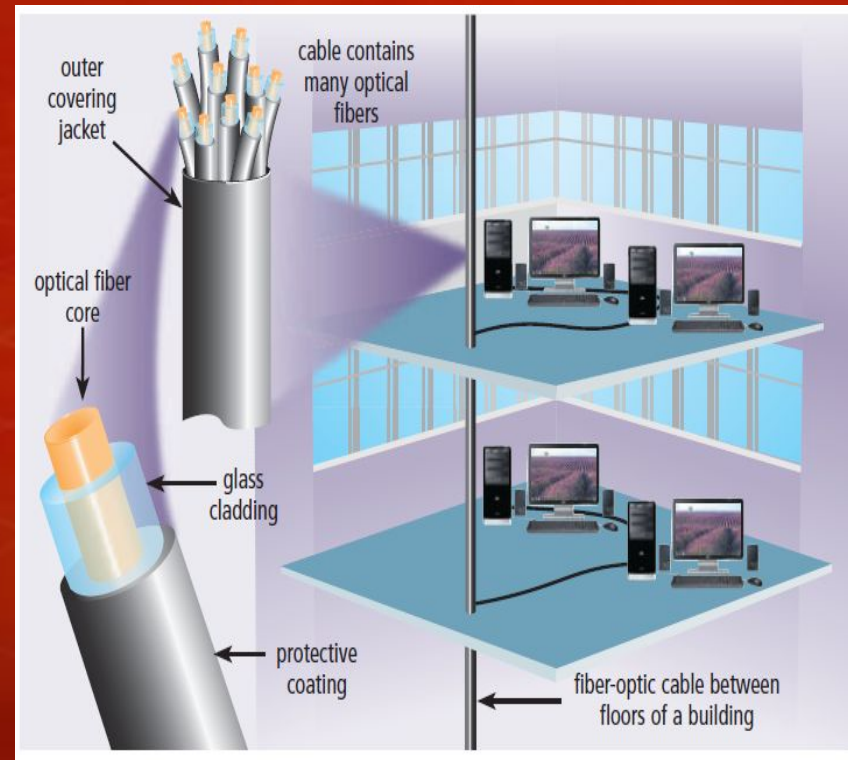
# Coaxial Cable

- ◆ Single copper core surrounded by insulation, metal shield, and outer cover.
- ◆ Used in cable TV and Internet connections.
- ◆ Advantages: better noise resistance than twisted pair.



# Fiber Optic Cable

- ◆ Transmits data using light pulses.
- ◆ Made of glass or plastic fibers.
- ◆ Advantages:
  - Very high bandwidth
  - Immune to electromagnetic interference
  - Long-distance capability



# Comparison: Wired Media

| Type         | Material        | Speed     | Distance | Cost   | Use Case                   |
|--------------|-----------------|-----------|----------|--------|----------------------------|
| Twisted Pair | Copper          | Medium    | Short    | Low    | LANs, telephones           |
| Coaxial      | Copper + shield | High      | Medium   | Medium | Cable TV, broadband        |
| Fiber Optic  | Glass / Plastic | Very High | Long     | High   | Internet backbone, telecom |

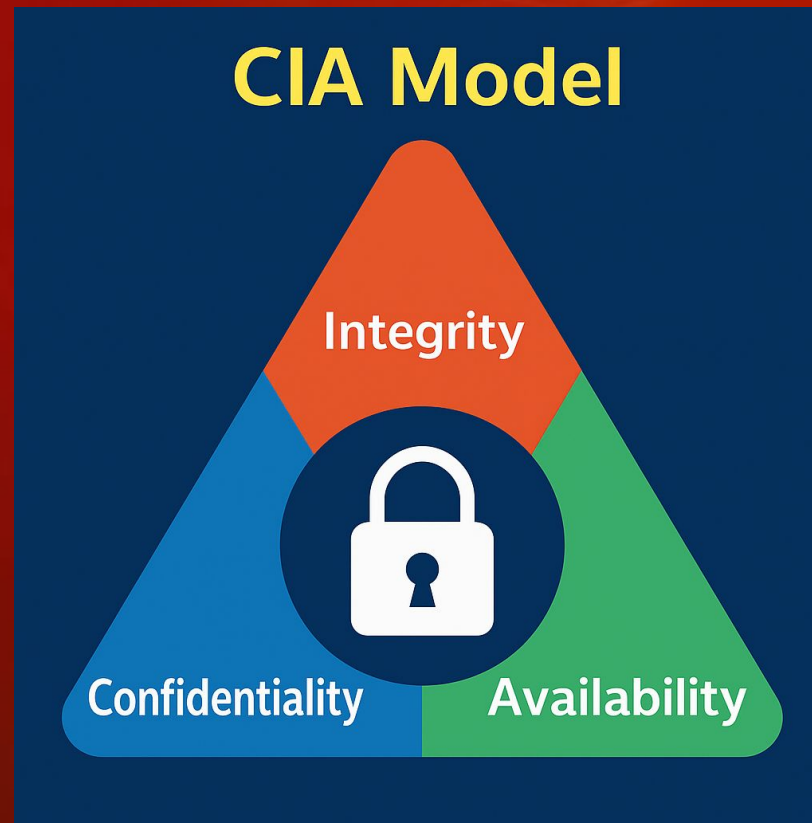
# Microwaves

- ◆ Used in satellite and cellular networks.
- ◆ Travel in straight lines; need line-of-sight.
- ◆ Faster than radio waves.



# What is Network Security?

- Refers to protecting data, devices, and networks from unauthorized access or attacks.
- Ensures Confidentiality, Integrity, and Availability (CIA) of information.





# CIA Triad

- ◆ **Confidentiality** means that sensitive information stays in the right hands, blocking unauthorized access.
- ◆ **Integrity** is what keeps data accurate and consistent, preventing tampering or unintended changes during storage or transmission.
- ◆ **Availability** assures network services and data are accessible when authorized users need them.



# CIA Summary

| Term            | Focus             | Main Threat         | How to Protect             |
|-----------------|-------------------|---------------------|----------------------------|
| Confidentiality | Keep secrets safe | Unauthorized access | Encryption, authentication |
| Integrity       | Keep data correct | Data alteration     | Hashing, permissions       |
| Availability    | Keep data usable  | Downtime, attacks   | Backups, redundancy        |

# Common Security Threats

| Threat Type  | Description                    | Example              |
|--------------|--------------------------------|----------------------|
| Virus        | Infects files and replicates   | Infected attachments |
| Worm         | Self-replicating over networks | Email-spreading worm |
| Trojan Horse | Disguised as safe software     | Fake antivirus       |
| Spyware      | Monitors user activity         | Keylogger            |
| Phishing     | Tricking users to share info   | Fake bank email      |
| Ransomware   | Locks data until payment       | WannaCry attack      |

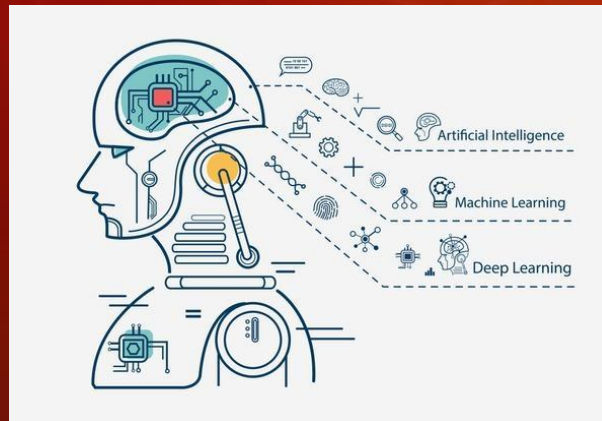
# Encryption Concept

- ◆ Converts data into unreadable form (cipher text).
- ◆ Only authorized users can decrypt it.
- ◆ Example:
  - Plain text → Encryption → Cipher text → Decryption → Plain text.

| Stage                             | Message | Readable?                                                                                 |
|-----------------------------------|---------|-------------------------------------------------------------------------------------------|
| Plain Text                        | HELLO   |  Yes |
| After Encryption<br>(Cipher Text) | JGNNQ   |  No  |
| After Decryption                  | HELLO   |  Yes |

# What is Artificial Intelligence (AI)?

- AI = Ability of machines to think, learn, and act like humans
- Uses data to solve problems and make decisions
- Examples → Siri answers your questions, Google suggests routes



# Examples of AI in Daily Life

## Area

Navigation

Shopping

Entertainment

Healthcare

Education

## Example of AI Use

Google Maps traffic updates

Amazon product recommendations

Netflix / YouTube suggestions

Disease detection / robotic surgery

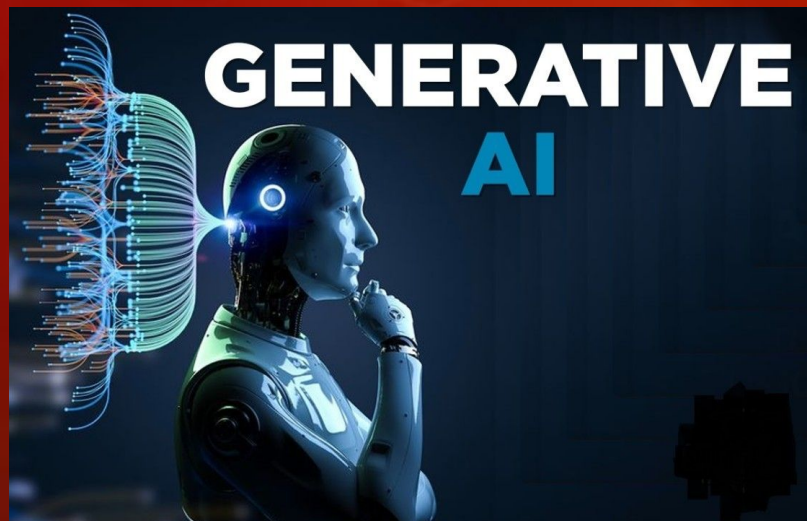
Smart learning apps and AI tutors

# Types of AI

| Type                   | Description                  | Example                     |
|------------------------|------------------------------|-----------------------------|
| Narrow AI (Weak AI)    | Works for one specific task  | Siri, Google Search         |
| General AI (Strong AI) | Thinks and acts like a human | Research robots / future AI |

# What is Generative AI?

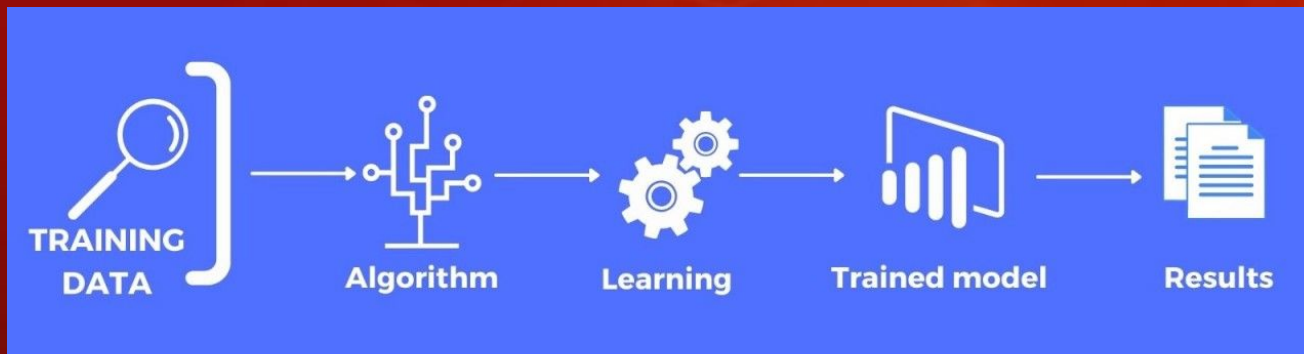
- ◆ Generative AI = AI that creates new content
- ◆ Uses machine learning models to generate text, images, music, or video
- ◆ Works by learning patterns from large datasets





# How Generative AI Works

- ◆ **Data Collection** – AI learns from millions of examples
- ◆ **Training** – Finds patterns and relationships
- ◆ **Generation** – Produces new text, image, or code based on input prompt





# Examples of Generative AI Tools

| Tool Name            | Type              | Output Example             |
|----------------------|-------------------|----------------------------|
| ChatGPT              | Text generation   | Essays, explanations, code |
| DALL·E / Leonardo AI | Image generation  | Drawings, digital art      |
| Synthesia            | Video generation  | AI avatars & lectures      |
| Soundful             | Audio generation  | AI music                   |
| Runway ML            | Video & animation | Editing, creative tools    |

# What is a Prompt?

- ◆ A prompt is a text instruction you give to an AI model
- ◆ The quality of your prompt determines the quality of the result
- ◆  **Example:**
  - **Prompt:** “Write a 50-word paragraph about climate change in simple English.”
  - **Output:** (AI-generated paragraph)

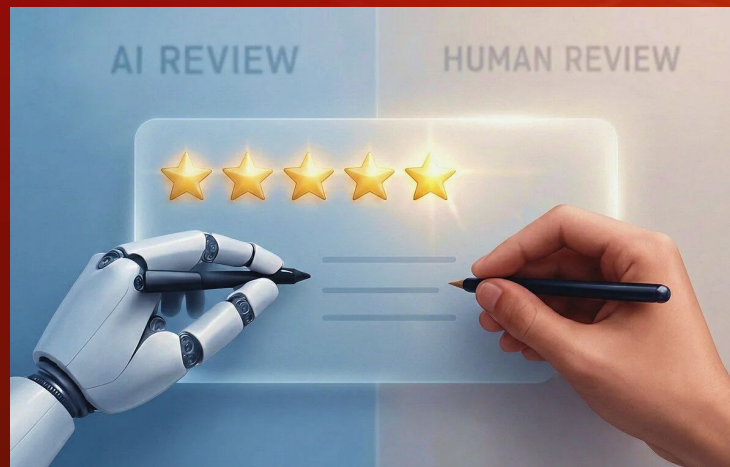


# How to Write Good Prompts

- ◆ Be clear and specific
- ◆ Give context (what, who, how)
- ◆ Ask for format or style (paragraph, list, code, etc.)
- ◆ Test and refine your prompt
- ◆ Example Comparison:
  - Bad Prompt → “Explain AI”
  - Better Prompt → “Explain AI to a 12-year-old student with one example.”

# Ethical and Responsible Use

- Always verify information from AI tools
- Avoid plagiarism and misinformation
- Respect privacy and intellectual property
- Use AI as an assistant, not a replacement for thinking



# How Blockchain Uses Networks

## Aspect

## Role of Network

### Communication

Nodes share transaction data across the network.

### Verification

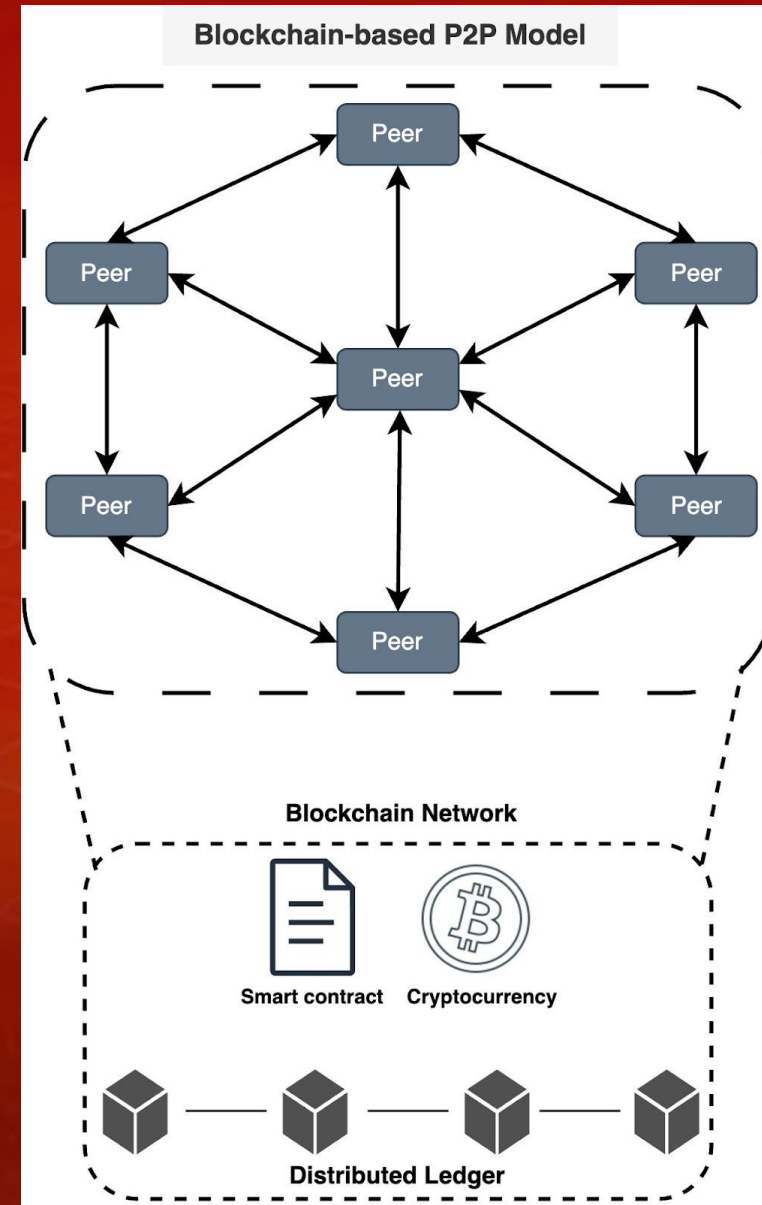
Distributed computers (peers) validate transactions together.

### Storage

Data is stored across many devices (no single central server).

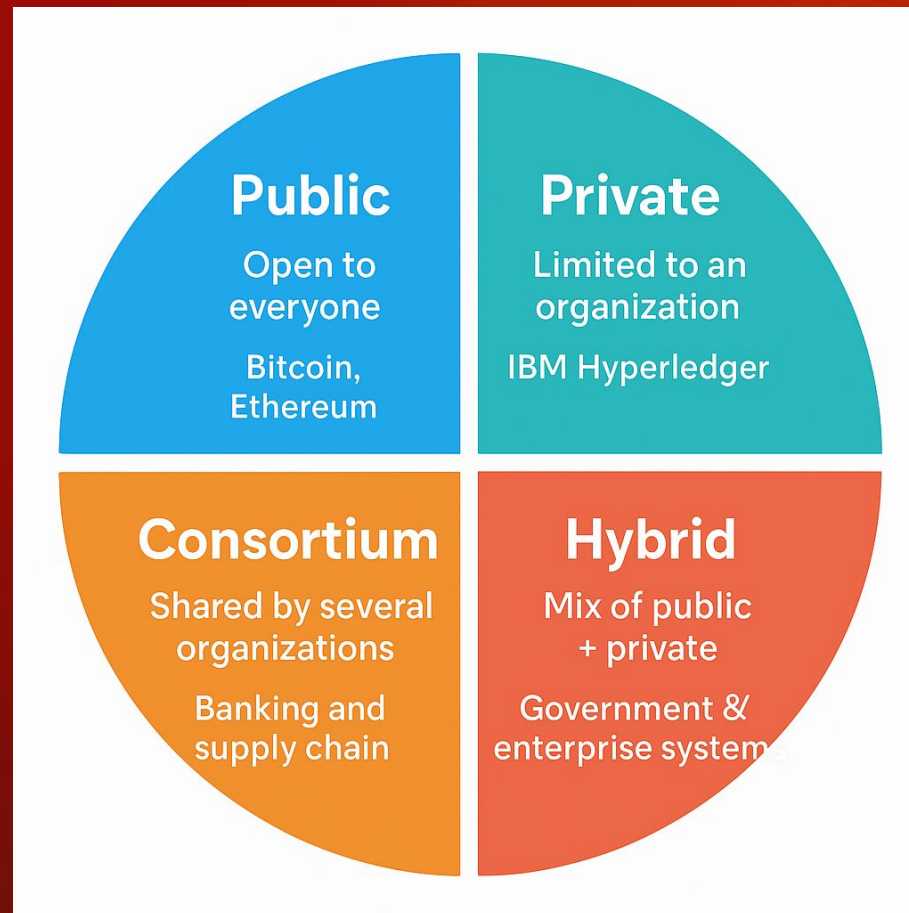
### Security

Cryptographic linking and consensus make tampering almost impossible.



# Types of Blockchain Networks

Different blockchain networks provide varying levels of control, transparency, and decentralization.





# Applications of Blockchain

- ◆ Cryptocurrency: Bitcoin, Ethereum
- ◆ Smart Contracts: Automated digital agreements
- ◆ Supply Chain Tracking: Ensures authenticity of goods
- ◆ Healthcare: Secure patient data sharing
- ◆ Voting Systems: Transparent and tamper-proof
- ◆ Digital Identity: Protects user privacy





# Summary

- ◆ Revise for Final

# Suggested Reading

- ◆ All my lectures and recommended book chapters